

Data sheet for the Capstone Project

1. Dataset motivation

The dataset was created to support the optimisation of eight benchmark black-box objective functions. Each observation represents a candidate solution and the resulting objective score, allowing surrogate models to approximate the response surface.

2. Data composition

Data inputs

Inputs are structured in the format:

$$[X_1, X_2, X_3, \dots, X_i]$$

Where the number of input variables i correlates to the dimensionality, or number of input features, of the respective objective function. Inputs were given scaled with values between 0 and 1.

Data outputs

Outputs are represented as y , and vary in sign and magnitude between objective functions. This is the output score given by the black-box model for a given set of input values.

3. Data collection process

Initial datasets for each objective function were provided at the beginning of the project to allow for an initial approximation of the objective functions to be made. Any new data points that were added during the project came from interactions with the black-box model through weekly submissions, with inputs being chosen by the user and outputs being the resulting objective value.

4. Data preprocessing

Input values were naturally scaled to contain values between 0 and 1, and output values were naturally manipulated to transform each of the objective functions into a maximisation problem.

During the project, some of the output values were normalised in order to improve surrogate performance. No missing values were present.

5. Uses

The dataset was used to train surrogate models that approximate the black-box objective functions. These models support acquisition functions such as UCB (Upper Confidence Bound) and EI (Expected Improvement) when selecting future query points.

6. Distribution

Access

Data is stored within the project repository and is available to instructors and project viewers.

Limitations

The dataset is relatively small due to the initial number of provided data points being low, and because new evaluations are expensive, being limited to just 13 new data points for each function through the entirety of the project. Additionally, data may be biased towards local high-performing regions due to Bayesian Optimisation's exploitative behaviour combined with the limited number of queries which does not allow for exploring the entire search space.

7. Maintenance

The dataset is updated after each submission, with the user-chosen inputs and the resulting outputs being appended to the previous dataset. It is maintained alongside the codebase to provide traceability and allow for the development of the dataset to be as transparent as possible and ensure reproducibility of results.

8. Overview

Although relatively small, the dataset provides the information necessary for sample-efficient optimisation of the black-box objective. By recording both candidate solutions and objective values, it enables surrogate modelling and informed query selection while maintaining a manageable data collection cost.